

Experiment 2: Bayes Decision Theory

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Predicted class for new instance: benign
(base) bhanu@Bhanus-MacBook-Air C01-ASS3 % python Q2.py
Message: Congratulations! You won a free lottery. Claim now
Prediction: spam
Posterior Probabilities: [[0.03151112 0.96848888]]
❖ (base) bhanu@Bhanus-MacBook-Air C01-ASS3 %
```

Result

The SMS Spam Collection Dataset was successfully loaded and converted into numerical features using CountVectorizer. A Multinomial Naive Bayes classifier was trained using the dataset.

For the test message:

"Congratulations! You won a free lottery. Claim now."

the classifier predicted the message as **Spam** with a high posterior probability.

Thus, Bayes Decision Theory successfully classified the SMS message into the correct category.